

02 — Small Battery 300 kW (budget exhausts mid-cascade)

Proves · What happens when the budget runs out mid-cascade: propulsion takes all 300, the hotel row gets nothing.

Mechanics this scenario turns on

- Cascade per row: H = what it wants · $I = \min(\text{remaining budget}, H)$ · $J = I \times \text{CoverageFactor}$ (covered) · $L = H - J$ (left to the gensets). Priority: DpReserve → DpDemand → Mission → Propulsion → Hotel. Invariant $\Sigma J + \Sigma L = \Sigma H$ — the sea sets the swing, the battery moves the split.
- Peak-shaving H = average $\times$ VariationFactor (propulsion 5 %, hotel 2 %). CoverageFactor is a modelling assumption about how much of a swing a battery can realistically catch — propulsion 0.35, hotel 0.05. Both live in `appsettings.json`, not in code.
- Past saturation extra battery power buys nothing: once ΣH is fully taken, the remaining budget has no swing left to cover.

Panels described below · Battery Contribution · Power Demands · Baseline & IL1 · Battery Benefit

Anything not described here behaves exactly as in `01-excel-baseline` — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 01.

Same vessel as 01; only battery power 1260 → **300**.

Battery Contribution

Row	H	I	J	L	Budget after
Propulsion	573.15	$\min(300, 573.15) = \mathbf{300}$	$300 \times 0.35 = \mathbf{105}$	468.15	0
Hotel	76	$\min(0, 76) = \mathbf{0}$	0	76	0

Tiles: **PS = 105 · SR = 544.15**. Invariant check: $105 + 544.15 = 649.15$ — same total swing as 01; the smaller battery only moved the split toward the gensets.

Power Demands

Propulsion' = $11\,463 + 468.15 = \mathbf{11\,931}$ · Hotel' = $3\,800 + 76 = \mathbf{3\,876}$ (hotel got nothing!) SG 3 250 · AE **626** · ME = **15 181** (63.25 %) · Energy 75 905 750 / 3 130 000.

Baseline & IL1

Same 5 combos at slightly higher loads: optimal 2.67423 → IL1 **13 371.2 t/yr**; baseline (3rd-from-worst) 2.67850 → **13 392.5**; savings **21.4 t = \$16 664**.

Battery Benefit

World B is IDENTICAL to test 01's (the battery doesn't exist there): 2.69198 t/h. Benefit = $(2.69198 - 2.67423) \times 5000 = \mathbf{88.74 \text{ t/yr} = \$69\,218}$.

Cross-check with 01: IL1 rose by $13\,371.2 - 13\,286.2 = 85.0 \text{ t}$, and the benefit fell by $173.66 - 88.74 = 84.9 \text{ t}$ — the fuel you stop saving is exactly the fuel you burn extra. Consistency ✓

Takeaway: priority matters (propulsion took everything, hotel starved), and battery value has diminishing returns — the first kW of budget is worth the most.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-10 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)